

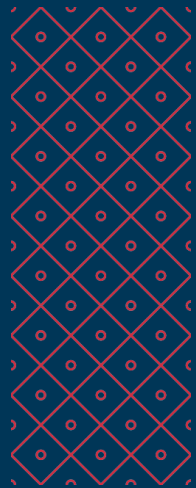


**Charles
University**

Programming in C++

Labs - winter 2025/26

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Lab 9

28/11/2025

Credit Project

Merge-request with detailed project description
by **Friday December 12th**

- E.g., 1-2 pages **README.md**
- Describe used **libraries**: what they can/cannot do
- The programmed **functionality**
- The programmed **functionality** from user perspective

Homework Discussion

Templates

```
class complex {  
    int real_; int im_;  
public:  
    int &re() { return real_; }  
    int re() const { return real_; };  
    int &im() { return im_; };  
    int im() const { return im_; };  
};  
  
void print(int v) { cout << v; }  
void print(double v) { cout << v; }  
void print(const string &str) { cout << str; }  
  
int main() {  
    complex c;  
    print(c.re());  
}
```



```
template<typename T>  
class complex {  
    T real_; T im_;  
public:  
    T &re() { return real_; }  
    T re() const { return real_; };  
    T &im() { return im_; };  
    T im() const { return im_; };  
};  
  
template<class T>  
void print(const T &v) { cout << v; }  
  
int main() {  
    complex<int> c;  
    complex<double> c;  
    print(c.re());  
}
```

Equivalent to using auto:
void print(const auto &v)

Static instantiation:
Create two different types

Templates

class = template

```
template<class Cont>
class container_wrapper {
    Cont cont;
```

Can we somehow
restrict allowed types?
E.g., no `complex<string>`

```
public:
    void push_back(
        const typename Cont::value_type& val) {
        cont.push_back(val);
    }

    void push_back(typename Cont::value_type &&rval) {
        cont.push_back(std::move(rval));
    }
};
```

A type or a static member?
→ Hint the compiler

```
template<class Cont>
auto make_wrapper() {
    return container_holder<Cont>();
}
```

```
auto vector_wrapper =
    make_wrapper<std::vector<int>>>();
vector_wrapper.push_back(3);
```

Concepts*

```
#include <concepts>
```

satisfied iff T is
an integral type

```
template<std::integral T>
class Complex {
    T real_; T im_;
public:
    T &re() { return real_; }
    T re() const { return real_; };
    T &im() { return im_; };
    T im() const { return im_; };
};
```

Specify a new
concept

Can be combined with basic bool operators

```
template<class T>
concept UnsignedIntegral =
    std::integral<T>
    && !std::signed_integral<T>;
```

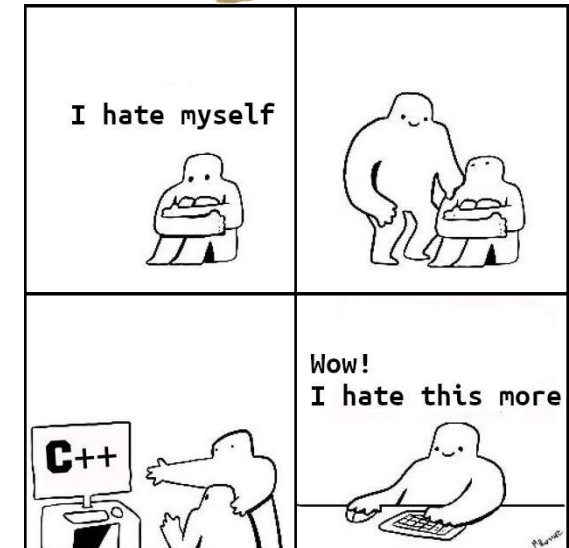
Some other examples

```
// a type is derived from another type
std::derived_from<Derived, Base>
// an object of a type can be moved
std::movable<T>
...
```

Concepts** (Advanced C++)

```
template<class T>
struct IsIntegral : std::bool_constant<
    // T* parameter excludes reference types
    requires (T t, T* p, void (*f)(T))
{
    // Exclude class types
    reinterpret_cast<T>(t);
    // Exclude enumeration types
    f(0);
    // Exclude everything but integral types
    p + t;
}> {};
```

```
template<class T>
concept Integral = IsIntegral<T>::value;
```



Work: Change your containers to support any type

Polymorphic vector (lab 8)

vector<T> (lab 6)

linked_list<T> (lab 6)

Matrix3D (lab 4)

Bonus: Improve performance of polymorphic vector (→ 1p)

Insert in $O(1)$ in amortized

Add benchmark measurements (insert + iteration)

Add missing operations

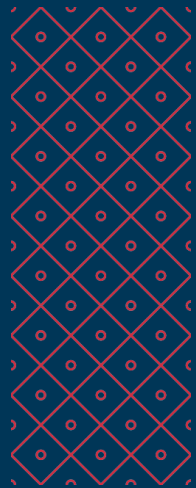
push_back, insert, pop_back, erase

Hints

Use templates

Submit the final version to Gitlab

- Each container → 1 point (4 in total)
- Create a *merge-request*
- Deadline: Thursday December 4th 5:00



Lab 8

21/11/2025

Credit Project

Merge-request with detailed project description
by **Friday December 12th**

- E.g., 1-2 pages **README.md**
- Describe used **libraries**: what they can/cannot do
- The programmed **functionality**
- The programmed **functionality** from user perspective
- I'll put an example and recommendation on my website

Survey Discussion

Big thank you!

- 1. More specific assignments**
2. Labs on work with files
3. Be more in sync with the lectures
4. ReCodex tests
5. Less granular points, ping-pong the solutions
6. Too much time spent on the homeworks

Survey Discussion

Big thank you!

- 1. More specific assignments**
2. Labs on work with files
3. Be more in sync with the lectures
4. ReCodex tests
5. Less granular points, ping-pong the solutions
6. Too much time spent on the homeworks

Survey Discussion

Big thank you!

More specific assignments

Motivation

real-world

I'll try to make it more specific

Email/DM me with the specific questions – I'm the stakeholder

Comment with how you understood it is fine

Survey Discussion

Big thank you!

Labs on work with files

There is going to be one in 2 weeks

Survey Discussion

Big thank you!

Be more in sync with the lectures

Motivation

Get quickly hands on *important* stuff

Here the things multiple times from different people

I'll try to be more in sync

Survey Discussion

Big thank you!

Use ReCodex

Motivation

Real-world (testing is part of the job)

I (personally) don't like Recodex: hard to debug failures

MR approach better for students

Survey Discussion

Big thank you!

Less granular points, possibility to ping-pong the solutions

Motivation

We already can ping-pong using MR

Points are less granular on purpose to motivate fix issues

Survey Discussion

Big thank you!

Too much time spent on the assignments

Motivation

Prepare you for the practical exam

I'll try to reevaluate the complexity of the assignments

(The deadline is not always strictly enforced)

Use AI to help you with the search/code

- *Always understand the generated code, don't just copy*

Homework Discussion

I'll go through the merge-requests (new and old) by end of the weekend

Inheritance

To inherit (=have access to) attributes and methods of the ancestor class

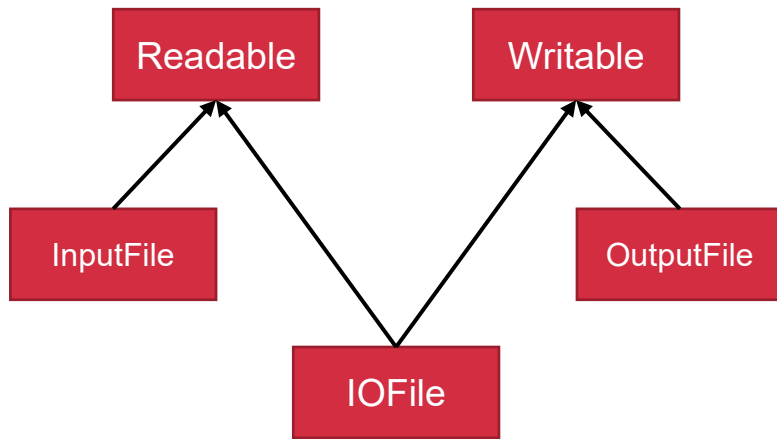
Interface implementation

Typically use of abstract classes

```
class Readable {};  
class InputFile : public Readable {};
```

```
class Writable {};  
class OutputFile : public Writable {};
```

```
class IOFile: public Writable, public Readable {};
```

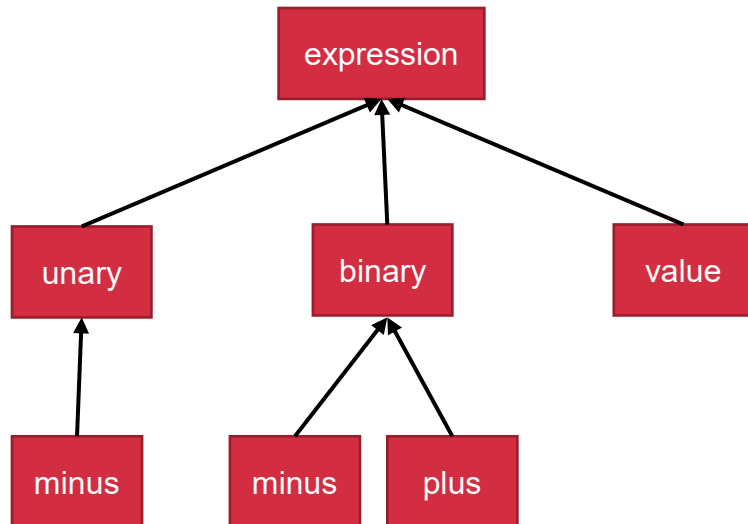


Inheritance

To inherit (=have access to) attributes and methods of the ancestor class

IS-A hierarchy

Might inherit some common attributes



```
class expression {};  
class binary : public expression {};  
class plus : public binary {};
```

```
class car {};  
class volvo : public car {};  
class skoda : public car {};
```

```
class animal {};  
class dog : public animal {};  
class cat : public animal {};
```

```
class employee {};  
class accountant : public employee {};
```

```
class object {};  
class triangle : public object {};  
class polygon : public object {};
```

Inheritance: Rule-of-Thumbs

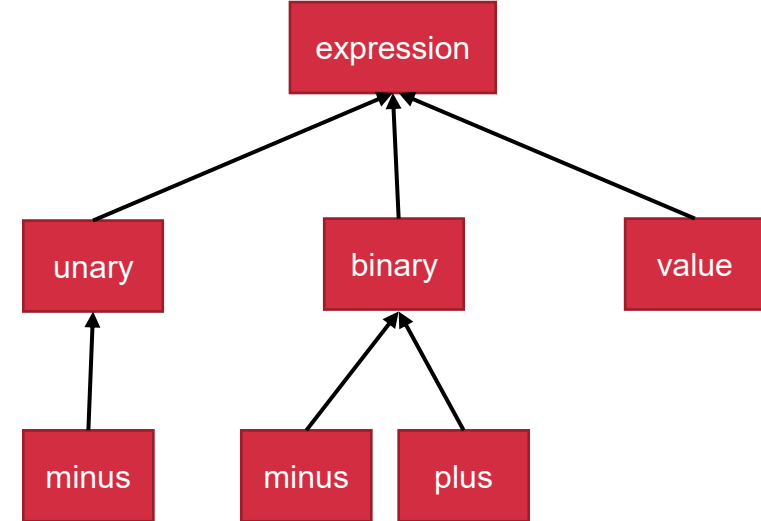
Think about the API

Other inheritance usage typically means bad design

if-else → bad OOP design



Don't forget to
break in switch



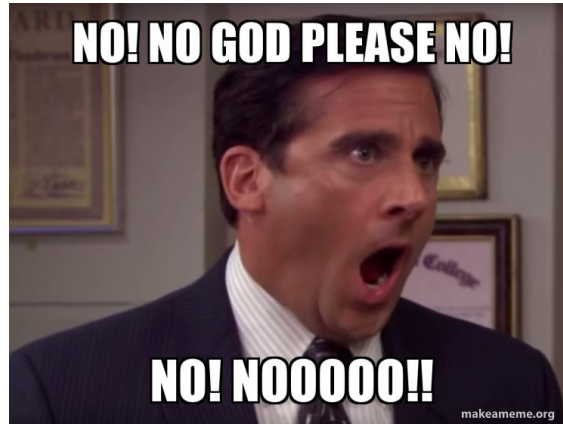
```
switch (type) {  
    case ExprType::MINUS:  
        // something  
        break;  
    case ExprType::PLUS:  
        // something else  
        break;  
    ...  
}
```

Inheritance: Rule-of-Thumbs

Think about the API

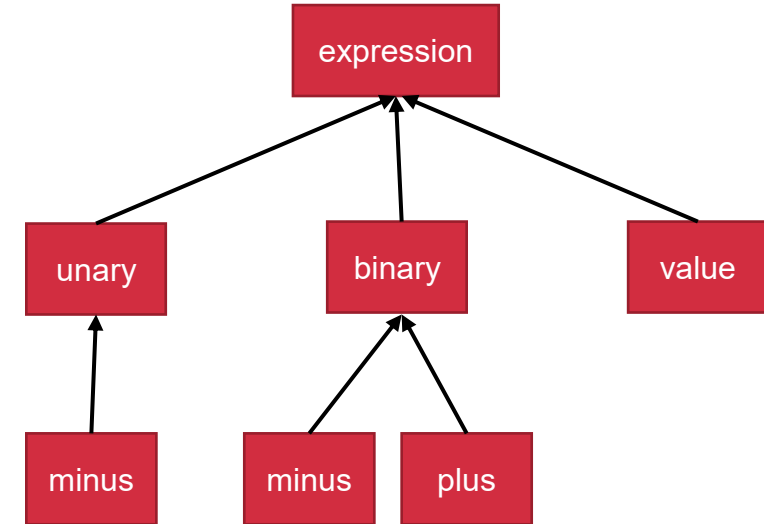
Other inheritance usage typically means bad design

if-else → bad OOP design



Don't forget to
break in switch

What to do about it?



```
switch (type) {  
    case ExprType::MINUS:  
        // something  
        break;  
    case ExprType::PLUS:  
        // something else  
        break;  
    ...  
}
```


Polymorphism

Base class acts as the (internally stored)
derived class

Function must be **virtual** and gets called via
reference/pointer

```
int main() {  
    derived d;  
    d.print(); // derived: 0  
    base &b = d;  
    b.print(); // derived: 0  
    base *b_ptr = &d;  
    b_ptr->print(); // derived: 0  
    base b2 = d;  
    b2.print(); // base: 0  
}
```

```
class base {  
protected:  
    int value = 0;  
public:  
    virtual ~base() = default;  
    virtual void print() const {  
        std::cout << "base: " << value;  
    }  
};  
  
class derived : public base {  
public:  
    void print() const override {  
        std::cout << "derived: " << value;  
    }  
};
```

Work 1: Polymorphic Vector

Create a vector which can store different types
(int, double, string)

Use inheritance (no union, variant, ...)

Hints

Create a wrapper for inserted types

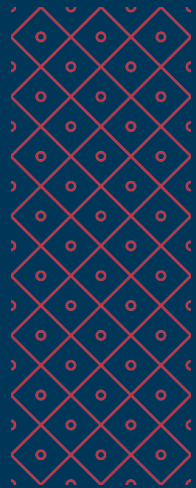
Use overloading

Submit the final version to Gitlab → 1 point

- **Create a merge-request**
- **Deadline: Thursday November 27th 5:00**

```
// API
// insert a value
push_back(value);
// print the value of i-th element
print(size_type i);
// print all values inside the array
print_all();
```

```
int main() {
    my_vec v;
    v.push_back(1);
    v.push_back(2.3);
    v.push_back("four");
    v.print(0); // 1
    v.print(2); // "four"
    v.print_all(); // [1, 2.3, "four"]
}
```



Lab 7

14/11/2025

Feedback

<https://forms.microsoft.com/e/bx4Tj8VW8J>

Course Evaluation Survey



Credit Project

Topic must be **approved** by Friday November 14th

- DM me Mattermost

Next

Approved Merge-request with detailed project description by **Friday December 12th**

Homework Discussion

```
vector(const vector &obj) :  
    capacity(obj.capacity),  
    size(obj.size)  
{  
    array_ptr = make_unique<int[]>(capacity);  
    for (std::size_t i = 0; i < size; ++i) {  
        array_ptr[i] = obj.array_ptr[i];  
    }  
}
```

```
vector & operator=(const vector &obj) {  
    if (this == &obj) {  
        return *this;  
    }  
    capacity = obj.capacity;  
    size = obj.size;  
    array_ptr = make_unique<int[]>(capacity);  
    for (std::size_t i = 0; i < size; ++i) {  
        array_ptr[i] = obj.array_ptr[i];  
    }  
    return *this;  
}  
  
vector & operator=(vector &&obj) {  
    capacity = obj.capacity;  
    size = obj.size;  
    array_ptr = std::move(obj.array_ptr);  
    obj.capacity = 0;  
    obj.size = 0;  
    return *this;  
}
```

Homework Discussion

```
vector(const vector &obj) :  
    capacity(obj.capacity),  
    size(obj.size)  
{  
    array_ptr = make_unique<int[]>(capacity);  
    for (std::size_t i = 0; i < size; ++i) {  
        array_ptr[i] = obj.array_ptr[i];  
    }  
}
```

DRY: Do not repeat yourself

Same code → function

```
vector &operator=(const vector &obj) {  
    if (this == &obj) {  
        return *this;  
    }  
    capacity = obj.capacity;  
    size = obj.size;  
    array_ptr = make_unique<int[]>(capacity);  
    for (std::size_t i = 0; i < size; ++i) {  
        array_ptr[i] = obj.array_ptr[i];  
    }  
    return *this;  
}  
  
vector &operator=(vector &&obj) {  
    capacity = obj.capacity;  
    size = obj.size;  
    array_ptr = std::move(obj.array_ptr);  
    obj.capacity = 0;  
    obj.size = 0;  
    return *this;  
}
```

Homework Discussion

```
vector(const vector &obj) :  
    capacity(obj.capacity),  
    size(obj.size)  
{  
    array_ptr = make_unique<int[]>(capacity);  
    for (std::size_t i = 0; i < size; ++i) {  
        array_ptr[i] = obj.array_ptr[i];  
    }  
}  
  
friend void swap(vector &lv, vector &rh) noexcept {  
    std::swap(lv.capacity, rh.capacity);  
    std::swap(lv.size, rh.size);  
    std::swap(lv.array_ptr, rh.array_ptr);  
}
```

Copy-and-swap idiom

Strong exception
guarantee

```
vector &operator=(const vector &obj) {  
    // ???  
}  
  
vector(vector &&obj) : vector{} {  
    // ???  
}  
  
vector &operator=(vector &&obj) {  
    // ???  
}
```

How to implement
using swap?

Homework Discussion

```
vector(const vector &obj) :  
    capacity(obj.capacity),  
    size(obj.size)  
{  
    array_ptr = make_unique<int[]>(capacity);  
    for (std::size_t i = 0; i < size; ++i) {  
        array_ptr[i] = obj.array_ptr[i];  
    }  
}
```

```
friend void swap(vector &lv, vector &rh) noexcept {  
    std::swap(lv.capacity, rh.capacity);  
    std::swap(lv.size, rh.size);  
    std::swap(lv.array_ptr, rh.array_ptr);  
}
```

Copy-and-swap idiom

Useful in concurrent
programming

```
vector &operator=(const vector &obj) {  
    vector tmp(obj);  
    swap(*this, tmp);  
    return *this;  
}
```

```
vector(vector &&obj) : vector{} {  
    swap(*this, obj);  
}
```

```
vector &operator=(vector &&obj) {  
    vector tmp(std::move(obj));  
    swap(*this, obj);  
    return *this;  
}
```

Homework Discussion

```
vector() : vector_capacity(8), vector_size(0) {  
    array_ptr = std::make_unique<int[]>(vector_capacity);  
}
```

How about this?

INIT_CAPACITY

```
vector(std::size_t size, int value) : vector_size(size) {  
    std::size_t cap = 8;  
    while (cap < size) {  
        cap *= 2;  
    }  
    vector_capacity = cap;  
    array_ptr = std::make_unique<int[]>(vector_capacity);  
    for (std::size_t i = 0; i < vector_size; ++i) {  
        array_ptr[i] = value;  
    }  
}
```

INIT_CAPACITY

CAPACITY_MULT

Function as Parameter

`std::function`

Functors

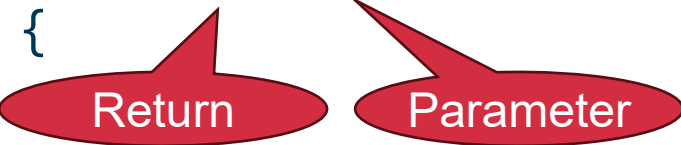
Lambdas

Templates

Function pointers/references (☹ C-style)

`std::function<RET(PAR1, PAR2,...)>`

```
void for_each(const std::vector<int> &vi,
             const std::function<void(int)> &fn) {
    for (int x : vi) {
        fn(x);
    }
}
```



The diagram shows two red speech bubble-like shapes. The first, labeled 'Return', points to the `void` return type of the `fn` parameter in the function signature. The second, labeled 'Parameter', points to the `int` argument `x` inside the `for` loop.

```
void print(int x) { cout << x << endl; }
using myfn_t = std::function<void(int)>;
```

```
int main(int argc, char* argv[]) {
    vector<int> vi{1, 2, 3, 4, 5, 6};
    myfn_t f = print;
    ...
    for_each(f, vi);
}
```

Function as Parameter

std::function

Functors

Lambdas

Templates

Function pointers/reference (C-style)

Can look anyway
you need

A class with operator()

```
struct PrintFtor {  
    void operator()(int x) const {  
        cout << x << endl;  
    }  
};
```

Operator()

Why is const here?

```
void for_each(const std::vector<int> &vi,  
             const PrintFtor &fn) {  
    for (int x : vi) {  
        fn(x);  
    }  
}
```

```
int main(int argc, char* argv[]) {  
    vector<int> vi{1, 2, 3, 4, 5, 6};  
    PrintFtor print;  
    ...  
    for_each(vi, print);  
}
```

Function as Parameter

std::function

Functors

Lambdas

Templates

Function pointers/references (☹ C-style)

```
void for_each(const std::vector<int> &vi,
              NumOddFtor &fn) {
    for (int x : vi) {
        fn(x);
    }
}
```

A class with operator()

```
struct NumOddFtor {
    size_t num = 0;
    void operator()(int x){
        num += x % 2;
    }
};
```

Can hold a state

```
int main(int argc, char* argv[]) {
    vector<int> vi{1, 2, 3, 4, 5, 6};
    NumOddFtor num_odd;
    ...
    for_each(vi, num_odd);
    cout << num_odd.num << endl;
}
```

Function as Parameter

std::function

Functors

Lambdas

Templates

Function pointers/references (☹ C-style)

```
void for_each(const std::vector<int> &vi,
              auto fn) {
    for (int x : vi) {
        fn(x);
    }
}
```

Needs auto

An anonymous in-place functor

```
[capture](par1,...){ /* body */ };
```

Passes things from the outside

```
auto l = [&num_odd, TWO](int x){
    num_odd += x % TWO;
};
```

By reference

By copy

```
int main(int argc, char* argv[]) {
    vector<int> vi{1, 2, 3, 4, 5, 6};
```

```
    ...
    for_each(f, [](int x){
        cout << x << endl;
    });
}
```

Algorithms

```
#include <algorithm> (+ namespace std::ranges)
```

A library defining functions for variety of purposes

- **Batch:** `for_each`, ...
- **Search:** `find`, `find_first`, `all`, `any`, ...
- **Fold:** `fold_left`, `fold_right`, ...
- **Modifying sequence:** `copy`, `copy_if`, ...
- **swap, minimum/maximum:** `swap`, `max`, `min`, `minmax`, ...
- **Transformation and Generation:** `transform`, `replace`, `fill`, `generate`, ...
- **Removing and Order-changing:** `remove`, `reverse`, `rotate`
- **Partitioning and Sorting:** `partition`, `sort`, `is_sorted`
- **Binary search, Set, Heap:** `lower_bound`, `binary_search`, `set_union`, `includes`, `make_heap`, `push_heap`, ...
- **Numeric:** `iota`, `accumulate`, `reduce`, ...
- **Lexicographical comparison, Permutation, on Unitialized memory**

Define range through iterators

Anything iterator-like (iterators, pointers, ...)

Define action through function/functor/lambda

Anything function-like (functions, functors, ...)

Work 1: People big data database

Store all information found on your **ID** (name, age, address, ...)

Efficient **search** based on individual fields

Compute **statistics** for individual fields (min, max, avg, histogram)

Optimize for big data but fit into your main memory

Hints

Think about efficient data structures

Careful with the copies (owners vs. observers)

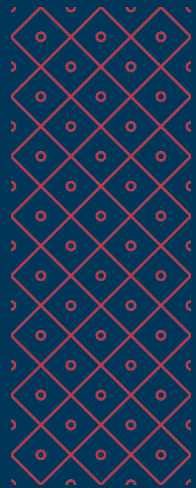
Do you need dynamic allocation?

Use algorithms

Submit the final version to Gitlab → 2 point

- **Create a *merge-request***
- **Deadline: Thursday November 20th 5:00**





Lab 6

07/11/2025

Credit Project

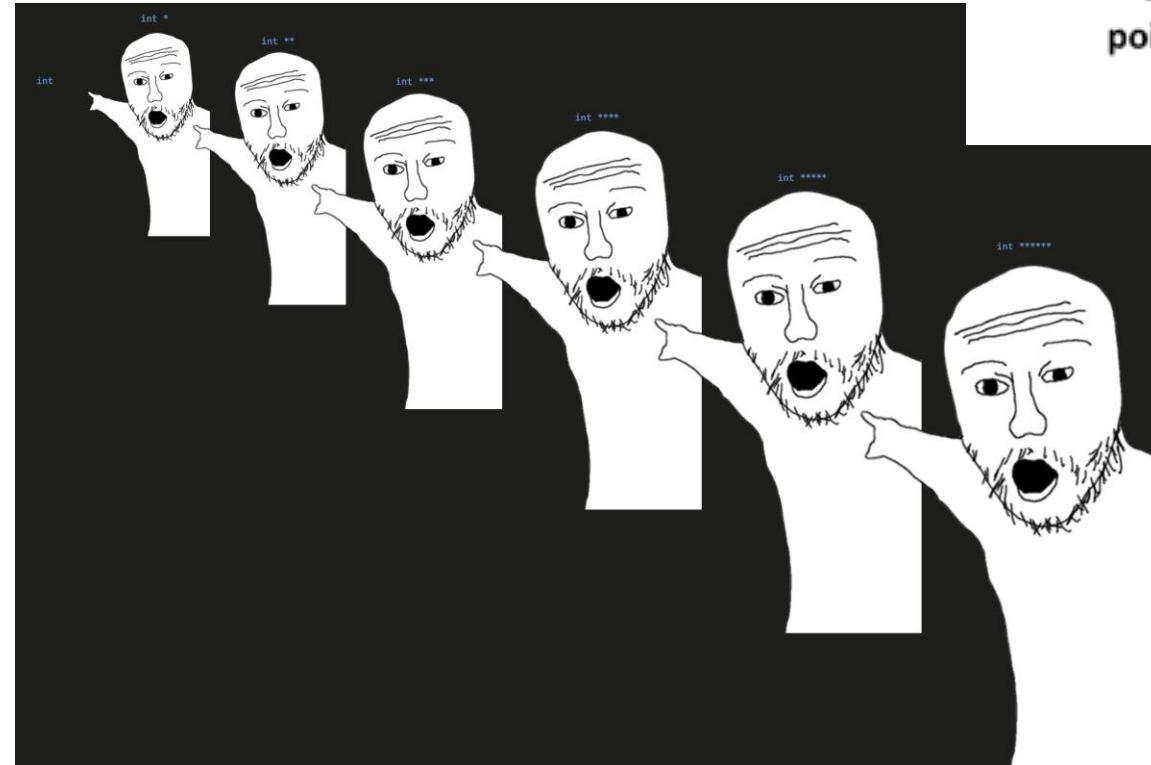
Topic must be **approved** by Friday November 14th

- DM me Mattermost

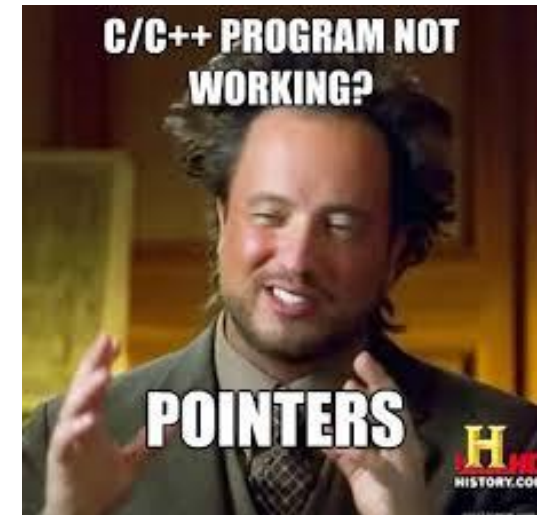
Homework Discussion

Pointers

random dev: I got stuck with this problem. Can anyone give me some pointers?



C++
raw pointers trying to prevent people from using raw pointers



Dynamic Allocation

Use smart pointers (no raw pointers, i.e., **no new/new[]**)

Single owner

Most common case

Passing the ownership - move only, no copy

`unique_ptr<T>`

Shared ownership (multiple owners)

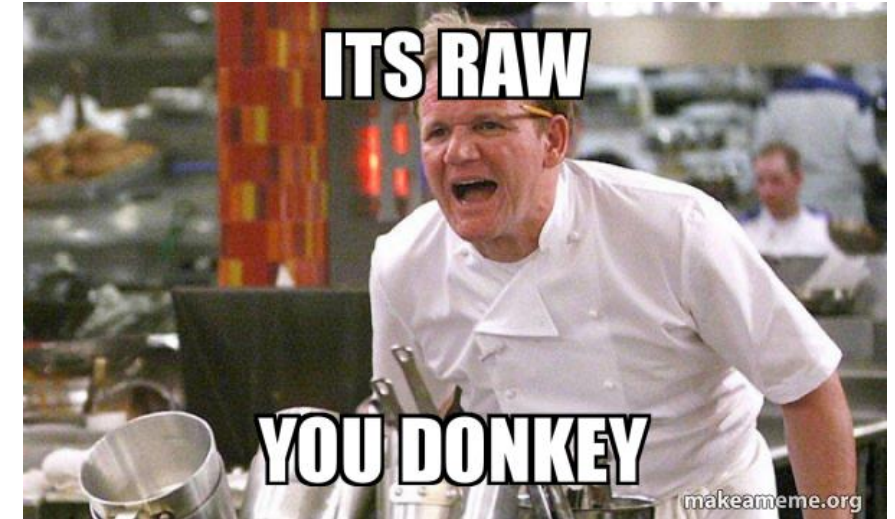
`shared_ptr<T>`

`weak_ptr<T>` // to break the cycle

Creation

`make_unique<T>`, `make_shared<T>`

`make_unique<int[10]>()` // Allocation of consecutive memory (~array)



Do you **really** need to own or just observe/work with the object?

Observers

Working with the pointer with **no changes to ownership**

Return type (pointer)

Modifying observer $\rightarrow T *$ (pointer)

Read-only observer $\rightarrow \text{const } T *$

Getting an observer

smart pointers $\rightarrow \text{get}()$

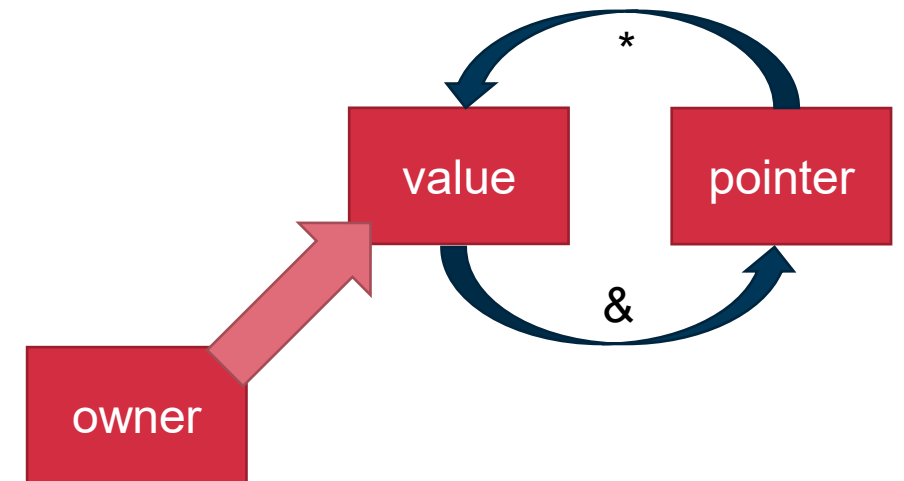
otherwise $\rightarrow \&$ (observer $\sim$ address)

Accessing the value

access the value directly $\rightarrow *$

access member attribute $\rightarrow ->$ (same as doing $(*\text{var})$).

Pointing to nothing $\rightarrow \text{nullptr}$



Dynamic Allocation: Final Notes

Prefer static/automatic storage

☺ `complex c;`

☹ `auto c = make_unique<complex>();`

Dynamic allocation
is slow!

Prefer containers

☺ `vector<int> vi(10);`

☹ `auto ia = make_unique<int[]>(10);`

Prefer `unique_ptr`

use `shared_ptr` only for complex ownership, e.g., unknown ownership protocol (multithreading)

Use only when necessary

Object lifetime doesn't correspond to function invocations

Polymorphism

Is it really that **complex**
or just **messy code**?

Pointers in Memory

```
int main() {  
    int i = 2;  
    int *pi = &i;  
    int **ppi = &pi;  
    cout << i;        // 2  
    cout << pi;       // ..00  
    cout << *pi;      // 2  
    cout << ppi;      // ..04  
    cout << *ppi;     // ..00  
    cout << **ppi;    // 2  
}
```

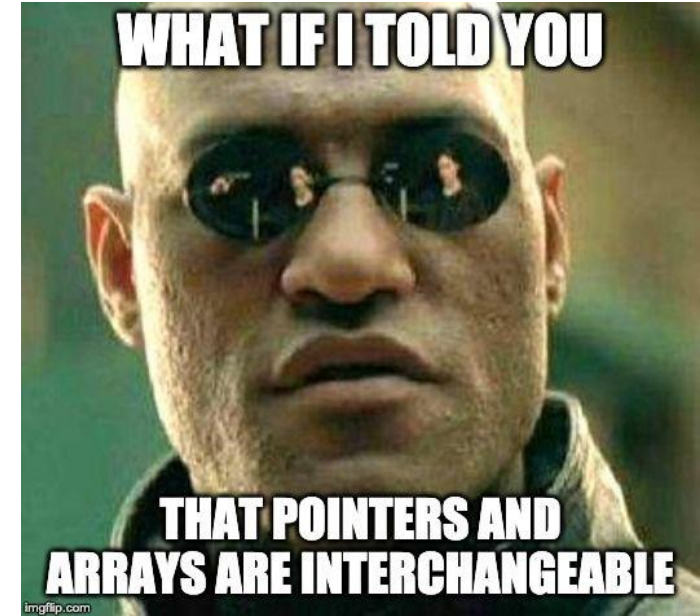
Address	Value	(Variable)	operator*
....			
..00	2	i	*
..02			
..04	..00	pi	*
..06			
..08	..04	ppi	*
..0a			
....			

What if we call
`cout << *i`

Pointer Arithmetic

```
unique_ptr<int[]> int_arr = make_unique<int[]>(10);
for (size_t i = 0; i < 10; ++i) { int_arr[i] = 0; }
int *ptr = int_arr.get();
(*ptr) = 1;
*(ptr + 1) = 2;
cout << *ptr << ' ' << *(ptr + 1) << endl; // 1 2
ptr += 1;
cout << *ptr++ << endl; // 2
cout << *ptr << ' ' << *(ptr - 1) << endl; // 0 2
*ptr-- = 3;
cout << ptr[-1] << ' ' << ptr[0] << ' ' << ptr[1] << endl; // 1 2 3
```

Takes type into account



Linked List: Example

```
struct node {
    unique_ptr<node> next;
    int value;

    node(int value, unique_ptr<node> &&next) :
        value(value), next(std::move(next)) {}
};

class linked_list {
    unique_ptr<node> first_node;
public:
    node *front() {
        return first_node.get(); // Observer
    }

    const node *back() const {
        node *ptr = first_node.get(); // Observer
        if (ptr != nullptr) {
            while (ptr->next != nullptr) {
                ptr = (*ptr).next.get(); // Equivalent to ->
            }
        }
        return ptr;
    }
};
```

```
void push_front(int value) {
    auto new_node = std::make_unique<node>(
        value,
        std::move(first_node));
    first_node = std::move(new_node);
}

void pop_front() {
    auto first = std::move(first_node);
    first_node = std::move(first->next);
} // automatic deallocation of first
```

Work 1: Finish the LL

`size()`, `print()`, `push_back()`, `pop_back()`
`ctor()`, `ctor(init_size, default_value)`, `dtor`
`operator[]`

Submit the final version to Gitlab → 1 point

- **Create a *merge-request***
- **Deadline: Thursday November 13th 5:00**

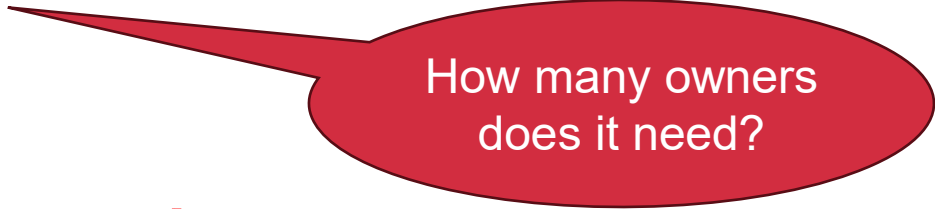
Work 2: vector<int>

Implement your own integer vector

Mandatory operations

default ctor, ctor(size_t, value_type), copy/move ctor/assignment
size(), capacity(), reserve(), push_back(), operator[]()

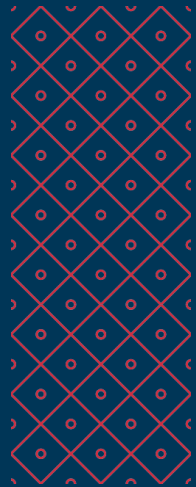
Use array allocation, no LL



How many owners
does it need?

Submit the final version to Gitlab → 2 point

- **Create a *merge-request***
- **Deadline: Thursday November 13th 5:00**



Lab 5

31/10/2025

Credit Project

Topic must be **approved** by Friday November 14th

- DM me Mattermost

Declaration/Definition

```
// file: my_class.hpp
```

```
#ifndef MY_CLASS_HPP  
#define MY_CLASS_HPP
```

```
void fn(int x);
```

```
class my_class {  
public:  
    my_class();  
    int exec(int x);
```

```
private:  
    double d;  
    static size_t i;  
};
```

```
#endif // MY_CLASS_HPP
```

```
// file: my_class.cpp
```

```
#include "my_class.hpp"  
#include <iostream>
```

```
void fn(int x) {  
    cout << "fn()";  
}
```

```
my_class::my_class() : d(1.0) {  
    cout << "ctor";  
}
```

```
int my_class::exec(int x) {  
    for(int i=0; i < x; ++i) { ... }  
}
```

```
size_t my_class::i = 0;
```

Declaration/Definition

```
// file: my_class.hpp
```

```
#ifndef MY_CLASS_HPP  
#define MY_CLASS_HPP
```

```
void fn(int x);
```

```
class my_class {  
public:  
    my_class();  
    int exec(int x);
```

Declaration

```
private:  
    double d;  
    static size_t i;  
};
```

```
#endif // MY_CLASS_HPP
```

```
// file: my_class.cpp
```

```
#include "my_class.hpp"  
#include <iostream>
```

Include header

```
void fn(int x) {  
    cout << "fn()";  
}
```

Definition

```
my_class::my_class() : d(1.0) {  
    cout << "ctor";  
}
```

```
int my_class::exec(int x) {  
    for(int i=0; i < x; ++i) { ... }  
}
```

```
size_t my_class::i = 0;
```


Declaration/Definition

// file: my_class.hpp

```
#ifndef MY_CLASS_HPP
#define MY_CLASS_HPP
```

```
void fn(int x);
```

```
class my_class {
public:
    my_class();
    int exec(int x);
```

```
private:
    double d;
    static size_t i;
};
```

```
#endif // MY_CLASS_HPP
```

Include guards

// file: my_class.cpp

```
#include "my_class.hpp"
#include <iostream>
```

```
void fn(int x) {
    cout << "fn()";
}
```

```
my_class::my_class() : d(1.0) {
    cout << "ctor";
}
```

```
int my_class::exec(int x) {
    for(int i=0; i < x; ++i) { ... }
}
```

```
size_t my_class::i = 0;
```

Declaration/Definition

// file: my_class.hpp

```
#ifndef MY_CLASS_HPP  
#define MY_CLASS_HPP
```

```
void fn(int x);
```

```
class my_class {  
public:  
    my_class();  
    int exec(int x);
```

```
private:  
    double d;  
    static size_t i;  
};
```

```
#endif // MY_CLASS_HPP
```

#pragma once

Include guards

// file: my_class.cpp

```
#include "my_class.hpp"  
#include <iostream>
```

```
void fn(int x) {  
    cout << "fn()";  
}
```

```
my_class::my_class() : d(1.0) {  
    cout << "ctor";  
}
```

```
int my_class::exec(int x) {  
    for(int i=0; i < x; ++i) { ... }  
}
```

```
size_t my_class::i = 0;
```

Sequence Containers

std::vector<Type> - dynamic array

- my_vec[idx] = value, push_back(), ...

std::array<Type, Size> - fixed size array

- my_array[idx] = value, ...

std::deque<Type> - double ended dynamic array

- push_back(), push_front(), back(), front(), ...

std::list<Type> - linked list

Adaptors

Stack, queue, priority_queue

flat_set, flat_map, flat_multiset, flat_multimap

Iterators

*An object representing **references to elements** inside a container*

```
struct Complex { double re; double im; };  
using ComplexVec = std::vector<ComplexVec>;  
ComplexVec cvec{{1., 1.}, {2., 2.}, {3., 3.}};  
  
for (ComplexVec::const_iterator i = cvec.cbegin(); i != cvec.cend(); i++)  
    cout << "[" << i->re << "," << i->im << "]" << endl;  
  
std::vector<int> vi{1, 2, 3};  
cout << "begin: " << *vi.begin() << ", end: " << *vi.end(); << endl;
```

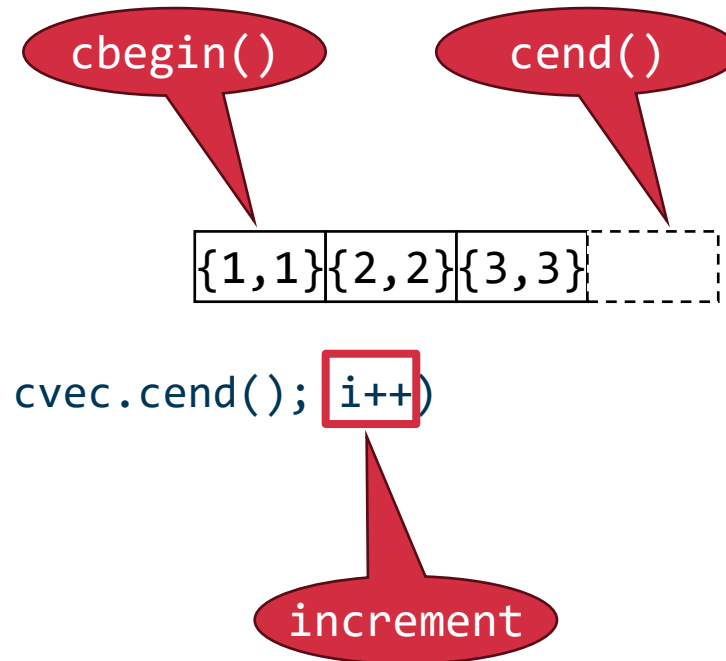
Iterators

An object representing **references to elements** inside a container

```
struct Complex { double re; double im; };  
using ComplexVec = std::vector<Complex>;  
ComplexVec cvec{{1., 1.}, {2., 2.}, {3., 3.}};
```

```
for (ComplexVec::const_iterator i = cvec.cbegin(); i != cvec.cend(); i++)  
    cout << "[" << i->re << "," << i->im << "]" << endl;
```

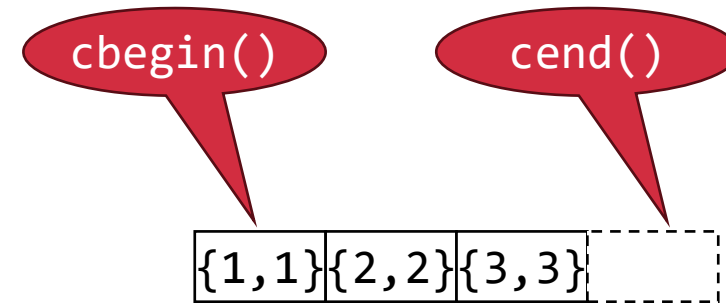
```
std::vector<int> vi{1, 2, 3};  
cout << "begin: " << *vi.begin() << endl;
```



Iterators

An object representing **references to elements** inside a container

```
struct Complex { double re; double im; };  
using ComplexVec = std::vector<ComplexVec>;  
ComplexVec cvec{{1., 1.}, {2., 2.}, {3., 3.}};
```



```
for (ComplexVec::const_iterator i = cvec.cbegin(); i != cvec.cend(); i++)  
    cout << "[" << i->re << "," << i->im << "]" << endl;
```

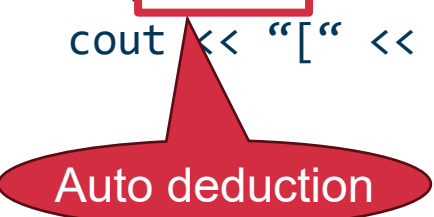
```
std::vector<int> vi{1, 2, 3};  
cout << "begin: " << *vi.begin() << ", end: " << *vi.end(); << endl;
```

What happens?

Iterators

*An object representing **references to elements** inside a container*

```
struct Complex { double re; double im; };  
using ComplexVec = std::vector<ComplexVec>;  
ComplexVec cvec{{1., 1.}, {2., 2.}, {3., 3.}};  
  
for (auto i = cvec.cbegin(); i != cvec.cend(); i++)  
    cout << "[" << i->re << "," << i->im << "]" << endl;
```



Auto deduction

Iterators

*An object representing **references to elements** inside a container*

```
struct Complex { double re; double im; };  
using ComplexVec = std::vector<ComplexVec>;  
auto cvec[{1., 1.}, {2., 2.}, {3., 3.}];  
  
for (auto i = cvec.cbegin(); i != cvec.cend(); i++)  
    cout << "[" << i->re << "," << i->im << "]" << endl;
```

What is the type here?

Work 1: Summing Program

Program that prints a sum of all numbers

Implement **special methods only in C/D**

- Ctors, dtors, operators, ...

Can add $O(1)$ **attributes into C** (e.g., no vector)

`C::print()` **only** used for printing

Goal: Understand **when** special methods are called

Submit the final version to Gitlab → 1 point

- **Deadline: Thursday November 6th 5:00**

```
class C {
    /* CAN ADD MORE ATTRIBUTES */
    const int value;
    /* USE THIS FUNCTION FOR PRINTING */
    void print() const {
        cout << value << "\n";
    }

public: /* IMPLEMENT SPECIAL METHODS ONLY */
};

class D {
    std::vector<C> cs; /* NO MORE ATTRIBUTES */
public: /* IMPLEMENT SPECIAL METHODS ONLY */
};

int main(int argc, char *argv[]) {
    int first, last;
    cin >> first >> last;
    cout << "Summing numbers:\n";
    D d(first, last); // prints: first, ..., last
    cout << "Preparing...\n";
    D d2 = d;
    cout << "Sum of the numbers:\n";
    d2 = d; // prints sum of numbers first...last
}
```

Work 2: N-in-row for P players

A generic N-in-row (“Piškvorky”) for P players

Set the **number of players** and **how many symbols in row** is needed

Set the **names** and **symbols** for individual players to **visualize** the game

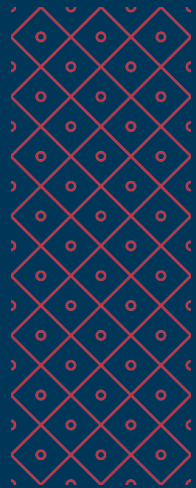
Players take **turns**

Game ends when the **first** player has **N-in-row**

Validate player **inputs**

Submit the final version to Gitlab → 3 point

- **Deadline: Thursday November 6th 5:00**



Lab 4

24/10/2025

Homework Feedback

Will be put today into Gitlab

The **first occurrence** of mistakes **can be corrected** during one iteration of code review.

Repeated mistakes will be **penalized**.

Defining Your Own Types

Use `using` (or `typedef` in old C/C++)

Can be templated (later)

```
using my_int = int;  
using int_pair_t = std::pair<my_int, my_int>;  
using my_string = std::vector<char>;  
using int_vector_t = std::vector<int>;
```

```
my_int x = 3;  
int_pair_t p{10, 20};  
my_string str = {'a', 'b', 'c'};  
int_vector_t vi(10, 0);
```

Constant Values

Read only value that cannot be changed

Naming values in code

~ Every number in the code should be a named constant

```
constexpr double PI = 3.14;
```

```
constexpr size_t MAX_SIZE = 16 * 1024 * 1024;
```

const

A constant value that cannot be changed

constexpr

Constant value (potentially) evaluated at

compile time

Can be used as arguments to templates

Both can be used together with `static` (later)

Argument Passing - Recap

How would you design a (global) function that sums two matrixes and return the new one?

Return

- a) `Matrix sum(...)`
- b) `const Matrix sum(...)`
- c) `Matrix &sum(...)`
- d) `const Matrix &sum(...)`
- e) `Matrix &&sum(...)`
- f) `Matrix *sum(...)`
- g) `const Matrix *sum(...)`

Argument Passing - Recap

How would you design a (global) function that sums two matrixes and return the new one?

Arguments

- a) `Matrix sum(Matrix m1, Matrix m2)`
- b) `Matrix sum(const Matrix m1, const Matrix m2)`
- c) `Matrix sum(const Matrix &m1, const Matrix &m2)`
- d) `Matrix sum(Matrix &m1, Matrix &m2)`
- e) `Matrix sum(Matrix &&m1, Matrix &&m2)`

Transferring Ownership

R-value references - &&

Moves the object somewhere (into a function)

- Use `std::move` on the caller side

The object no longer lives outside (the function)

Typical usage

A single owner, e.g., `std::unique_ptr`

Processing large objects

```
vector<unique_ptr<int>>::push_back(  
    unique_ptr<int> &&new_obj);
```

```
vector<unique_ptr<int>> vector_of_ints;  
vector_of_ints.push_back(move(make_unique<int>(x)));
```

Transferring Ownership

R-value references - &&

Moves the object somewhere (into a function)

- Use `std::move` on the caller side

The object no longer lives outside (the function)

Typical usage

A single owner, e.g., `std::unique_ptr`

Processing large objects

```
vector<unique_ptr<int>>::push_back(  
    unique_ptr<int> &&new_obj);
```

```
vector<unique_ptr<int>> vector_of_ints;  
vector_of_ints.push_back(move(make_unique<int>(x)));
```

```
vector<int> vi1{1, 2, 3};  
vector<int> vi2 = std::move(vi1);  
print(vi1);
```



What is inside vi1?

Special Methods

```
class Verbose {
    int x;
public:
    Verbose() {
        cout << "default ctor\n";
        this->x = 1;
    }

    Verbose(const Verbose &v) {
        cout << "copy ctor\n";
        this->x = v.x;
    }

    Verbose(Verbose &&v) {
        cout << "move ctor\n";
        this->x = v.x;
        v.x = 0;
    }

    ~Verbose() {
        cout << "dtor\n";
    }

    Verbose(int x) {
        cout << "user ctor\n";
        this->x = x;
    }
}
```

```
Verbose &operator=(const Verbose &v) {
    cout << "copy assignment\n";
    this->x = v.x;
    return *this;
}

Verbose &operator=(Verbose &&v) {
    cout << "move assignment\n";
    this->x = v.x;
    return *this;
}

};

int main()
{
    Verbose v1; // default ctor
    Verbose v2(2); // user ctor
    Verbose v3{3}; // user ctor
    Verbose v4(v2); // copy ctor
    Verbose v5 = v3; // copy ctor
    Verbose v6(std::move(v1)); // move ctor
    Verbose v7 = std::move(v4); // move ctor
    v1 = v2; // copy assignment
    v2 = std::move(v3); // move assignment
} // Calls destructors
```

Static Members

Attribute/method belongs to a class (not an object/instance)

Need to **share** attribute/method **among** all the objects/**instances**

Accesses through ::

Most things belong to an object

```
class Verbose {  
    static int cnt;  
}; // class
```

Declaration

```
int Verbose::cnt = 0;
```

Initialization

```
int main()  
{  
    Verbose v1; // 1st object/instance  
    Verbose v2(2); // 2nd object/instance  
    std::cout << Verbose::cnt;  
}
```

Access

Static Members - Example

```
class CountingClass {
    static size_t num_instances;

    static void change_num_instances(int val) {
        num_instances += val;
    }

public:
    static bool has_instance() {
        return num_instances > 0;
    }
    static size_t get_num_instances() {
        return num_instances;
    }

    CountingClass() { ??? }
    ~CountingClass() { ??? }

    CountingClass(const CountingClass &) {
        ???
    }
    CountingClass(CountingClass &&) { ??? }
};

size_t CountingClass::num_instances = 0;
```

```
void f() {
    cout << CountingClass::get_num_instances() << endl; // 0
    CountingClass cc1;
    cout << CountingClass::get_num_instances() << endl; // 1
    CountingClass cc2 = cc1;
    cout << CountingClass::get_num_instances() << endl; // 2
    std::vector<CountingClass> ccs(10);
    cout << CountingClass::get_num_instances() << endl; // 12
}

int main() {
    cout << CountingClass::get_num_instances() << endl; // 0
    f();
    cout << CountingClass::get_num_instances() << endl; // 0
}
```

Static Members - Example

```
class CountingClass {
    static size_t num_instances;

    static void change_num_instances(int val) {
        num_instances += val;
    }

public:
    static bool has_instance() {
        return num_instances > 0;
    }
    static size_t get_num_instances() {
        return num_instances;
    }

    CountingClass() { change_num_instances(+1); }
    ~CountingClass() { ??? }

    CountingClass(const CountingClass &) {
        ???
    }
    CountingClass(CountingClass &&) { ??? }
};

size_t CountingClass::num_instances = 0;
```

```
void f() {
    cout << CountingClass::get_num_instances() << endl; // 0
    CountingClass cc1;
    cout << CountingClass::get_num_instances() << endl; // 1
    CountingClass cc2 = cc1;
    cout << CountingClass::get_num_instances() << endl; // 2
    std::vector<CountingClass> ccs(10);
    cout << CountingClass::get_num_instances() << endl; // 12
}

int main() {
    cout << CountingClass::get_num_instances() << endl; // 0
    f();
    cout << CountingClass::get_num_instances() << endl; // 0
}
```

Static Members - Example

```
class CountingClass {
    static size_t num_instances;

    static void change_num_instances(int val) {
        num_instances += val;
    }

public:
    static bool has_instance() {
        return num_instances > 0;
    }
    static size_t get_num_instances() {
        return num_instances;
    }

    CountingClass() { change_num_instances(+1); }
    ~CountingClass() { change_num_instances(-1); }

    CountingClass(const CountingClass &) {
        ???
    }
    CountingClass(CountingClass &&) { ??? }
};

size_t CountingClass::num_instances = 0;
```

```
void f() {
    cout << CountingClass::get_num_instances() << endl; // 0
    CountingClass cc1;
    cout << CountingClass::get_num_instances() << endl; // 1
    CountingClass cc2 = cc1;
    cout << CountingClass::get_num_instances() << endl; // 2
    std::vector<CountingClass> ccs(10);
    cout << CountingClass::get_num_instances() << endl; // 12
}

int main() {
    cout << CountingClass::get_num_instances() << endl; // 0
    f();
    cout << CountingClass::get_num_instances() << endl; // 0
}
```

Static Members - Example

```
class CountingClass {
    static size_t num_instances;

    static void change_num_instances(int val) {
        num_instances += val;
    }

public:
    static bool has_instance() {
        return num_instances > 0;
    }
    static size_t get_num_instances() {
        return num_instances;
    }

    CountingClass() { change_num_instances(+1); }
    ~CountingClass() { change_num_instances(-1); }

    CountingClass(const CountingClass &) {
        change_num_instances(+1);
    }
    CountingClass(CountingClass &&) { ??? }
};

size_t CountingClass::num_instances = 0;
```

```
void f() {
    cout << CountingClass::get_num_instances() << endl; // 0
    CountingClass cc1;
    cout << CountingClass::get_num_instances() << endl; // 1
    CountingClass cc2 = cc1;
    cout << CountingClass::get_num_instances() << endl; // 2
    std::vector<CountingClass> ccs(10);
    cout << CountingClass::get_num_instances() << endl; // 12
}

int main() {
    cout << CountingClass::get_num_instances() << endl; // 0
    f();
    cout << CountingClass::get_num_instances() << endl; // 0
}
```


Static Members - Example

```
class CountingClass {
    static size_t num_instances;

    static void change_num_instances(int val) {
        num_instances += val;
    }

public:
    static bool has_instance() {
        return num_instances > 0;
    }
    static size_t get_num_instances() {
        return num_instances;
    }

    CountingClass() { change_num_instances(+1); }
    ~CountingClass() { change_num_instances(-1); }

    CountingClass(const CountingClass &) {
        change_num_instances(+1);
    }
    CountingClass(CountingClass &&) { ??? }
};

size_t CountingClass::num_instances = 0;
```

```
void f() {
    cout << CountingClass::get_num_instances() << endl; // 0
    CountingClass cc1;
    cout << CountingClass::get_num_instances() << endl; // 1
    CountingClass cc2 = cc1;
    cout << CountingClass::get_num_instances() << endl; // 2
    std::vector<CountingClass> ccs(10);
    cout << CountingClass::get_num_instances() << endl; // 12
}

int main() {
    cout << CountingClass::get_num_instances() << endl; // 0
    f();
    cout << CountingClass::get_num_instances() << endl; // 0
}
```

Try yourself

Work 1: Implement Class C

Implement class C so the program outputs:

1,2,3,...,16

Touch **only** class C, nothing else

- Nothing can be added/changed main() or fn_XXX()

Don't use exit(), break, goto, ...

Hint: which methods are called?

Submit the final version to Gitlab → 1 point

- **Deadline: Thursday October 30th 5:00**

```
class C { /* implement me */ };
```

```
// Don't touch anything below!!!
void fn_copy(C) {}
void fn_cref(const C&) {}
void fn_rref(C&&) {}

int main(int argc, char* argv[])
{
    cout << "1\n";
    C c1;
    cout << "3\n";
    C c2(c1);
    cout << "5\n";
    C c3 = c2;
    cout << "7\n";
    fn_copy(c1);
    cout << "9\n";
    fn_cref(c1);
    fn_copy(std::move(c1));
    fn_rref(std::move(c2));
    cout << "11\n";
    c3 = c2;
    cout << "13\n";
    c2 = std::move(c1);
    cout << "15\n";
}
```

Work 2: 3D Matrix for Integers v2.0

New API

- `print()`
- `sort_vector(x, y) // check STL`

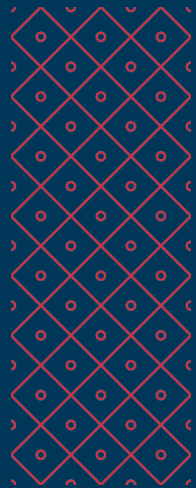
Try and benchmark performance of different containers: `std::vector`, `std::deque`, `std::list`, `std::array`

- Should be ~2 (un)commented lines only to change the container
- `std::chrono::system_clock::now();`
- Put your measured numbers into a comment
 - **Release** mode, **large enough tests**, ...

Correct all issues from the previous HW
Implement correctly all special methods
Show usage/test

Submit the final version to Gitlab → 2 points

- **Deadline: Thursday October 30th 5:00**



Lab 2

10/10/2025

Homework

Comments in Gitlab

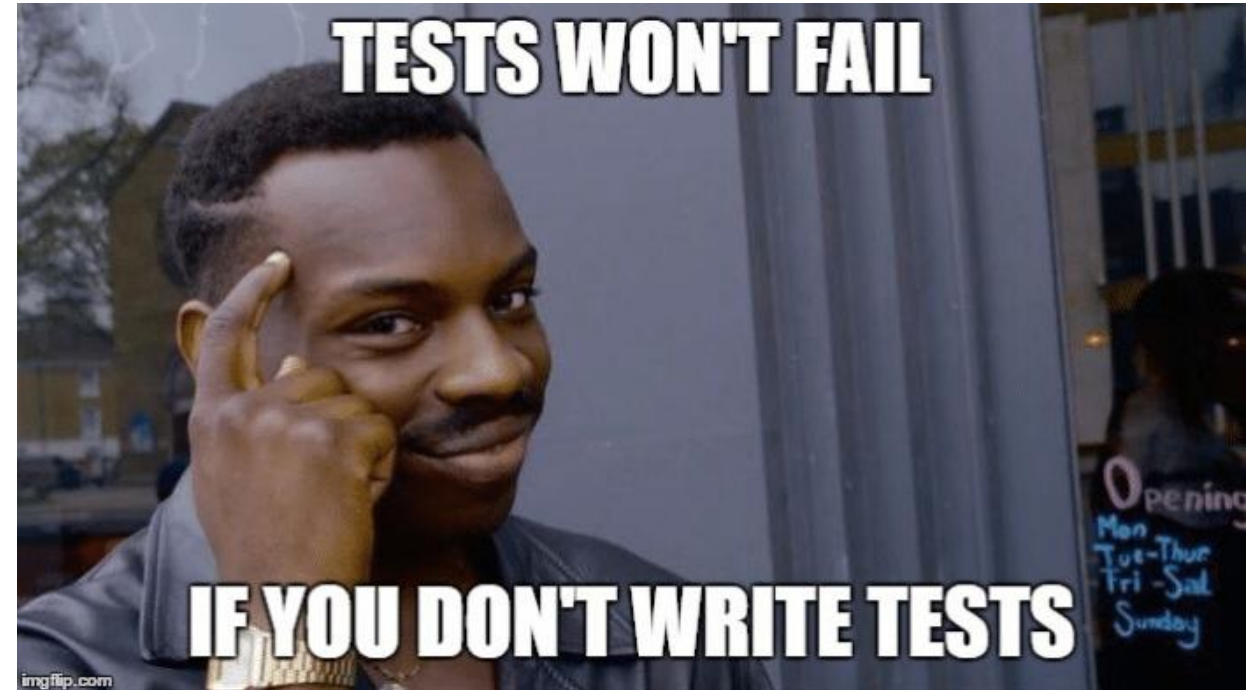
Points can be found in “Studijní mezivýsledky”

Homework Feedback

Testing is integral part of the assignment

I expected to see at least an attempt

Don't choose the easier way out



Homework Feedback

Testing is integral part of the assignment

Use STL wherever possible

- Someone **smart** already spent lots of time **programming** and **testing** it

`std::stod` instead of handmade parsing:

When I show my C code to

C Developers



ProgrammerHumor.io

C++ Developers



@12_Semitones

Core Guidelines

Set of guidelines for using C++ well

<https://isocpp.github.io/CppCoreGuidelines/CppCoreGuidelines>

<https://isocpp.github.io/CppCoreGuidelines/CppCoreGuidelines#fcall-parameter-passing>



Parameter Passing

Rule of thumb

Size	In	In-Out
Small	int	int &
Large	const string &	string &

```
_GLIBCXX_NODISCARD _GLIBCXX20_CONSTEXPR
reference
operator[](size_type __n) _GLIBCXX_NOEXCEPT
{
    __glibcxx_requires_subscript(__n);
    return *(this->_M_impl._M_start + __n);
}

/GLIBCXX_NODISCARD _GLIBCXX20_CONSTEXPR
const_reference
operator[](size_type __n) const _GLIBCXX_NOEXCEPT
{
    __glibcxx_requires_subscript(__n);
    return *(this->_M_impl._M_start + __n);
}
```

Link: https://github.com/gcc-mirror/gcc/blob/master/libstdc%2B%2B-v3/include/bits/stl_vector.h

Parameter Passing

Rule of thumb

Size	In	In-Out
Small	int	int &
Large	const string &	string &

```
template<typename _CharT, typename _Traits, typename _Alloc>
    _GLIBCXX20_CONSTEXPR
    void
    basic_string<_CharT, _Traits, _Alloc>::
    _M_assign(const basic_string& __str)
    {
        if (this != std::__addressof(__str))
        {
            const size_type __rsize = __str.length();
            const size_type __capacity = capacity();

            if (__rsize > __capacity)
            {
                size_type __new_capacity = __rsize;
                pointer __tmp = _M_create(__new_capacity, __capacity);
                _M_dispose();
                _M_data(__tmp);
                ...
            }
        }
    }
```

Parameter Passing

Rule of thumb

Size	In	In-Out
Small	int	int &
Large	const string &	string &

```
template<typename _CharT, typename _Traits, typename _Alloc>
    _GLIBCXX20_CONSTEXPR
    typename basic_string<_CharT, _Traits, _Alloc>::pointer
    basic_string<_CharT, _Traits, _Alloc>::
        _M_create(size_type& __capacity, size_type __old_capacity)
    {
        // _GLIBCXX_RESOLVE_LIB_DEFECTS
        // 83. String::npos vs. string::max_size()
        if (__capacity > max_size())
            std::__throw_length_error(__N("basic_string::_M_create"));

        // The below implements an exponential growth policy, necessary to
        // meet amortized linear time requirements of the library: see
        // http://gcc.gnu.org/ml/libstdc++/2001-07/msg00085.html.
        if (__capacity > __old_capacity && __capacity < 2 * __old_capacity)
        {
            __capacity = 2 * __old_capacity;
            // Never allocate a string bigger than max_size.
            if (__capacity > max_size())
                __capacity = max_size();
            ...
        }
    }
```

Parameter Passing

Rule of thumb

Size	In	In-Out
Small	int	int &
Large	const string &	string &

```
template<typename _Ch_type, class _Alloc, class _Rx_traits>
inline bool
regex_search(const _Ch_type* __s,
              match_results<const _Ch_type*, _Alloc>& __m,
              const basic_regex<_Ch_type, _Rx_traits>& __e
              regex_constants::match_flag_type __f,
              = regex_constants::match_default)
{ return regex_search(__s, __s + _Rx_traits::length(__s), __m,
__e, __f); }
```

Parameter Passing

Rule of thumb

Size	In	In-Out
Small	int	int &
Large	const string &	string &

```
template<typename _Ch_type, class _Alloc, class _Rx_traits>
inline bool
regex_search(const _Ch_type* __s,
              match_results<const _Ch_type*, _Alloc>& __m,
              const basic_regex<_Ch_type, _Rx_traits>& __e
              regex_constants::match_flag_type __f,
              = regex_constants::match_default)
{ return regex_search(__s, __s + _Rx_traits::length(__s), __m,
__e, __f); }
```

To call non-const
method

Parameter Passing

Rule of thumb

Size	In	In-Out
Small	int	int &
Large	const string &	string &

Why is there no Out?



Parameter Passing

Rule of thumb

Size	In	In-Out
Small	int	int &
Large	const string &	string &

Use return



Parameter Passing

Rule of thumb

Size	In	In-Out
Small	int	int &
Large	const string &	string &

What if there are multiple values?



Returning from Functions

Rule of thumb

```
template< class T >  
void minmax(std::initializer_list<T> ilist,  
            T& min, T& max);
```

Returning multiple values

Violates SOLID?

- single responsibility principle

Returning from Functions

Rule of thumb

Returning multiple values

Violates SOLID?

- single responsibility principle

Refactoring

```
template< class T >  
void minmax(std::initializer_list<T> ilist,  
           T& min, T& max);
```

→

```
template< class T >  
T min(std::initializer_list<T> ilist);
```

```
template< class T >  
T max(std::initializer_list<T> ilist);
```

Returning from Functions

Rule of thumb

Returning multiple values

Does not violate SOLID

Use `std::pair`/`std::tuple`

```
template< class T >  
void minmax(std::initializer_list<T> ilist,  
           T& min, T& max);
```

→

```
template< class T >  
std::pair<T, T> minmax(  
    std::initializer_list<T> ilist );
```



Class/Struct

Put all related things (data, functions) together

- Represents objects in OOP
- almost everything should belong to a class

No real difference except for default visibility, inheritance, ...

- **class** – by default everything **private**
- **struct** – by default everything **public**

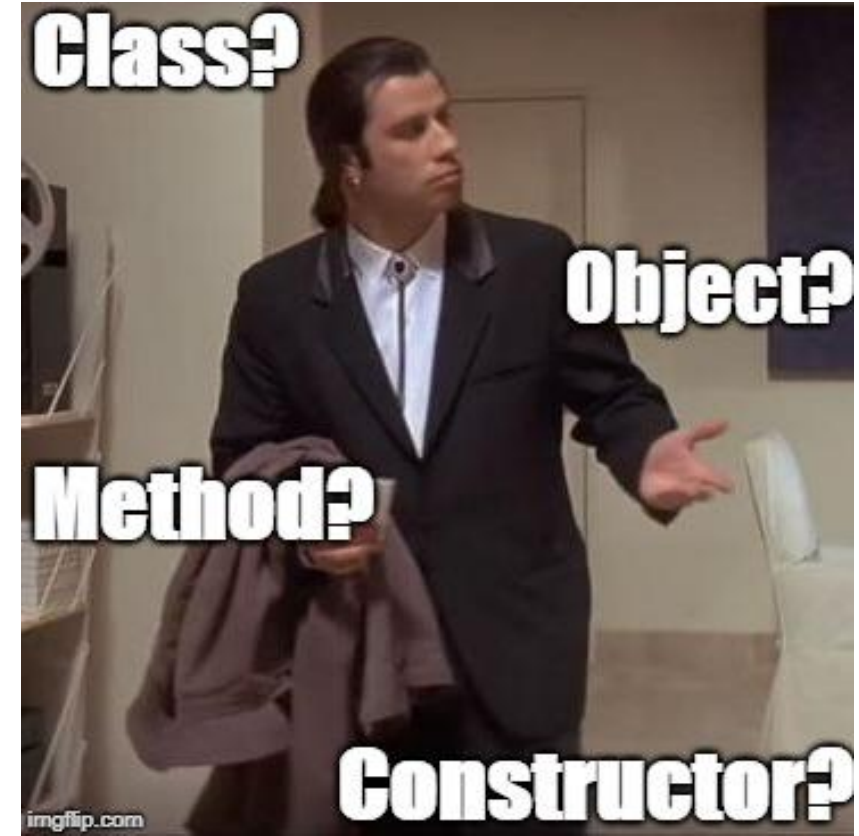
Internal things → **private**

- **protected** if need access from a child

Read-only functions → **const**

- const-correctness

Special methods (**constructor**, destructor, ...)



Class Example

```
class calculator {  
    // by default everything is private  
    void sum();  
    void subtract();  
  
public:  
    calculator() { /* default ctor */ }  
    calculator(const std::string &str) () {  
        /* ctor */  
    }  
    void calc(const std::string &str);  
    void print_result() const;  
  
private:  
    void multiply();  
  
protected:  
    void init();  
};
```

can be used
multiple times

semicolon
at the end!

```
calculator c; // no need for new  
c.calc("1+2-3");  
c.print_result();  
  
// calling non-default ctor  
calculator c2("1+2-3");  
c2.print_result();  
  
// creating a vector  
std::vector<calculator> calcs;
```

Class vs. Struct

Use `class` if the class has an invariant;

Use `struct` if the data members can vary independently

```
struct coordinate {  
    int x;  
    int y;  
    int z;  
  
    coordinate();  
    coordinate(int x);  
    coordinate(int x, int y);  
    coordinate(int x, int y, int z);  
  
    void set(int x, int y, int z);  
};
```

Dynamic Array - `std::vector<T>`

Beware of time complexity of individual methods

`vector<bool>` optimization

```
#include <vector>
int main() {
    std::vector<int> vi{1, 2, 3, 4, 5, 6};           // [1, 2, 3, 4, 5, 6]
    std::vector<float> vf(5, 0.0f);                 // [0.0, 0.0, 0.0, 0.0, 0.0]
    std::cout << vi[3] << " " << vf.at(3) << std::endl; // access the 4th! element
    std::cout << vi.size();
    vi[3] = 100; vi.at(6) = 600;                     // access the 4th and 7th element
    vf.push_back(100.0f); vf.emplace_back(200.0f);   // insert at the end
    vf.emplace_back(200.0f);                         // create element at the end
    vf.insert(3, 300.0f); vf.emplace(3, 300.0f);    // insert at the specific place
    vf.emplace(3, 300.0f);                          // create element at the specific place
    vi.pop_back();                                   // erase the last element
    vf.erase(2);                                    // erase the 3rd element
    vi.clear();                                      // clear whole container
    vi.reserve(10);                                 // reserve space(=memory) for 10 elements
    vi.resize(10);                                  // actually create 10 elements using default ctor
}
```

3D Matrix for Integers – minimal API

```
ctor(), ctor(x, y, z)
set(x, y, z, value), get(x, y, z), print()
set_width(), set_length(), set_height(), get_width(), get_length(),
get_height()
get_matrix(x), get_matrix(y), get_matrix(z)
get_vector(x, y), get_vector(y, z), get_vector(x, z)
clear() // set all values to 0 (zero)
fill_with_value(value) // set all values to a given value
num_zeros(), num_negatives(), num_positives(); // number of zeros/negatives/...
```

rename/adjust functions if needed – but keep the semantic

Submit the final version to Gitlab → 2 points

- **Deadline: Thursday October 16th 5:00**

3D Matrix for Integers - Hints

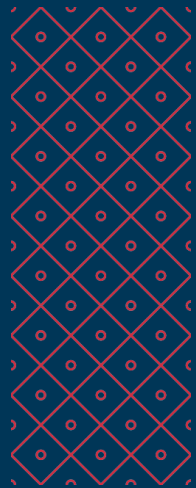
Think about the design

- Compose: array $\rightarrow$ matrix $\rightarrow$ 3-D matrix $\rightarrow$ 4D matrix $\rightarrow$... $\rightarrow$ XD matrix
- Design simple first, then continue to the next level

No need to focus too much on performance yet

Focus on:

- Passing arguments: const-references, references, ...
- const functions
- class design: decomposition into functions, function reusing, private/public, ...



Lab 1

3/10/2025



Basic information

Email: tomas.faltin@matfyz.cuni.cz

Labs web: <https://fan1x.github.io/teaching/2025-cpp>

Lecture web: <https://www.ksi.mff.cuni.cz/teaching/nprg041-web/>

Mattermost

- Invite link in [SIS/Notice-board](#)
- Channel: `2526/nprg041-cpp-faltin`
- DM: @faltin.tomas

Gitlab

- <https://gitlab.mff.cuni.cz/>
- <https://gitlab.mff.cuni.cz/teaching/nprg041/2025-26/klepl>



Labs Credit

Small (home) works assignments

- Submitted either to ReCodex or Gitlab
- At least 20 out of 30 points required
- Points contributed to the final exam grade

Credit project



Communication is the key

Don't be afraid to ask

Be proactive

- via email
- on Mattermost (instant)
 - DM if related to you only
 - Into a channel if others can benefit from it

If you struggle with something

If you feel like you might miss a deadline

Try things

Research project at our University

- Contact professor directly

Internships in companies

Erasmus

Conferences

...



Or not...

"There's a tiger in you"

Tiger in me :



MemeZilla.com

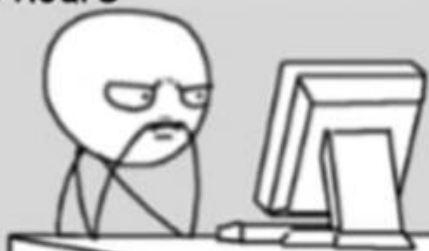


Let's get it started

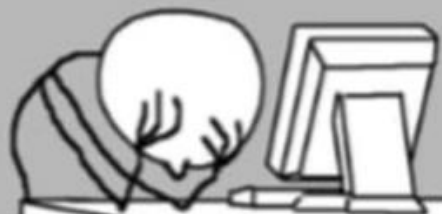
Motivation

Days before OpenAI

Developer coding
- 2 hours



Developer debugging
- 6 hours

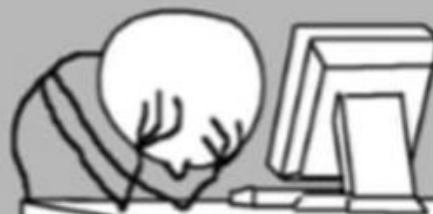


Days after OpenAI

ChatGPT generates
Codes - 5 min



Developer debugging
- 24 hours



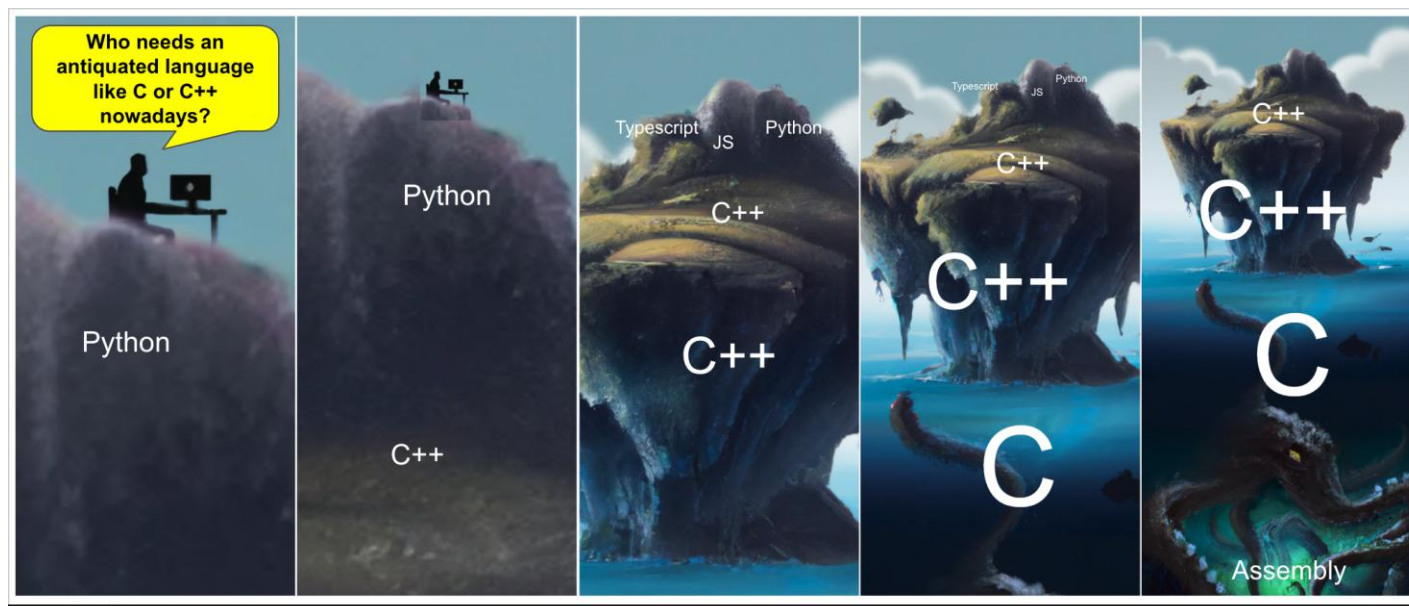
ProgrammerHumor.io



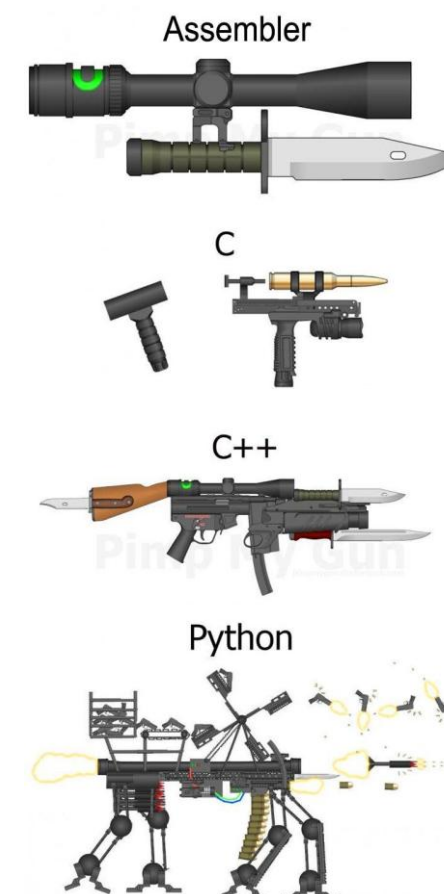
Why C++

“C makes it easy to shoot yourself in the foot. C++ makes it harder, but when you do, it blows away your whole leg.”

-- Bjarne Stroustrup



ProgrammerHumor.io





C++ (interesting) links

Reddit, Slack, ...

<https://en.cppreference.com/w/>

<http://www.cplusplus.com/>

<http://isocpp.github.io/CppCoreGuidelines/CppCoreGuidelines>

<https://www.youtube.com/user/CppCon>

<https://isocpp.org/>

<http://www.open-std.org/jtc1/sc22/wg21/docs/papers/>

<https://godbolt.org/>

...



Working Environment

Use anything you like
IDEs

- Visual Studio
 - License for students at <https://portal.azure.com/...>
- VS Code
- Clion
- Code::Blocks
- Eclipse
- ...

Compilers

- MSVC, GCC, Clang+LLVM, ICC, ...



Quick Recap



Code Requirements

Absence of Unforgivable Curses

- <https://teaching.ms.mff.cuni.cz/nswi170-web/pages/labs/coding/>

1. Code decomposition
2. Using constants
3. DRY
4. No global variables
5. Encapsulation

Code Requirements - Consistency

Consistency

- Be consistent within the code
- keep a single code style



Ministry Of Dev, PhD

@UdellGames

Folgen



Use whatever brace style you prefer.

But not this.

Don't do this.

Seek help instead of this.

```
public class Permuter
{
    private static void permute(int n, char[] a)
    {
        if (n == 0)
        {
            System.out.println(String.valueOf(a));
        }
        else
        {
            for (int i = 0; i <= n; i++)
            {
                permute(n-1, a);
                swap(a, n % 2 == 0 ? i : 0, n);
            }
        }
    }
    private static void swap(char[] a, int i, int j) {
        char saved = a[i];
        a[i] = a[j];
        a[j] = saved;
    }
}
```


Code Requirements – Readability

Code doesn't contain commented/dead parts
Code should be readable on its own
Comment complicated code



Code Requirements – Safe, Modern

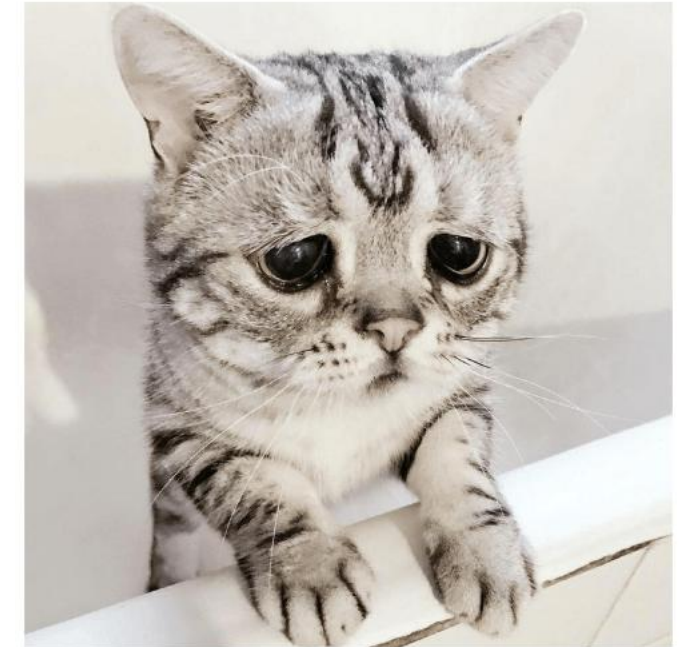
Prefer using modern constructs

Additional safety

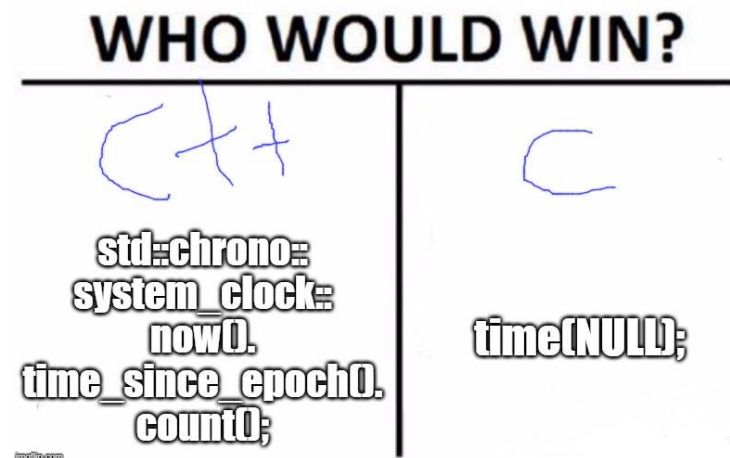
Maybe performance

E.g., prefer `std::vector<int>` to `new int[]`

Me when I realized that I can't pass 2D arrays to functions in C/C++ as `int a[]`:



"Pointers are a nuisance"



Code Requirements – Working

OFC, if the code is not working, all the above points are not that important they will help you with debugging at least 😊



Enough talking





Hello World

```
#include <iostream>
```

```
#include <string>
```

```
int main() {
```

```
    std::string name;
```

```
    std::cin >> name;
```

```
    std::cout << "Greetings from " << name << std::endl;
```

```
    return 0;
```

```
}
```

Hello World

```
#include <iostream>
#include <string>
```

```
int main() {
    std::string name;
    std::cin >> name;
    std::cout << "Greetings from " << name << std::endl;
    return 0;
}
```

Include the libraries
which implements the
used STL constructs
(string, cin, cout)

The main entry
point/function for all
programs. The
execution starts here

Declare a
variable of type
string

All the STL
constructs live
inside `std`
namespace

Write to
standard output
(screen)

Read from
standard input
(keyboard)



Compilation

c++ --version

- c++ is a compiler, here GCC

c++ hello_world.cpp -o hello_world

- Compile program into `hello\_world` executable (using default settings)

c++ -Wall -Wextra -Werror -O3 -std=c++2b hello_world.cpp -o hello_world

- Wall: Show all warnings
- Wextra: Show additional extra warnings
- Werror: Thread all warnings as errors
- O3: level of optimizations
- std=c++2b: Used C++ standard

Or use IDE ☺



More Complex Program

```
#include <iostream>
#include <string>
#include <vector>

using namespace std;

void pretty_print(const vector<string>& args) {
    // ... args[i]
}

int main(int argc, char** argv) {
    vector<string> args(argv, argv+argc);
    pretty_print(args);
    return 0;
}
```

More Complex Program

```
#include <iostream>
#include <string>
#include <vector>
```

Include the
whole std
namespace

```
using namespace std;
```

Passing the
argument by
(const)
reference

```
void pretty_print(const vector<string>& args) {
    // ... args[i]
}
```

Number
of
argument

Arguments of
the program on
the command
line

```
int main(int argc, char** argv) {
    vector<string> args(argv, argv+argc); // Wrap arguments
    pretty_print(args);
    return 0;
}
```

Transform
“magically” the
arguments into
C++ array of
strings

Functions And Parameters

```
int get_max(int v1, int v2) {  
    return v1 > v2 ? v1 : v2;  
}
```

```
int get_max1(const vector<int> &ints) {  
    int max = std::numeric_limits<int>::min();  
    for (int x : ints) {  
        max = get_max(x, max);  
    }  
    return max;  
}
```

```
bool get_max2(const vector<int> &ints, int &max) {  
    max = std::numeric_limits<int>::min();  
    for (int x : ints) {  
        max = get_max(x, max);  
    }  
    return !ints.empty();  
}
```

```
std::tuple<bool, int> get_max3(const vector<int> &ints) {  
    int max = std::numeric_limits<int>::min();  
    for (int x : ints) {  
        max = get_max(x, max);  
    }  
    return { !ints.empty(), max };  
}
```

Functions And Parameters

read-only input parameter

- Most of the types (string, vector, ...) → use const-reference - **const &**
 - `int get_max(const vector<int> &ints)`
- For small numeric types (int, float, double, ...) → use **direct parameter**
 - `int get_max(int v1, int v2)`

output parameters

- Single output parameter → use **return** value
 - `int get_max(const vector<int> &ints)`
- Few output parameters → use **tuple/pair/structure**
 - `std::tuple<bool, int> get_max(const vector<int> &ints)`
- Many output parameters → use reference - **&**
 - `bool get_max(const vector<int> &ints, int &max)`



Work

1. Hello World
2. A greeting program (use names from arguments)
 - ``hello.exe Adam Eve`` → ``Hello to Adam and Eve``
 - What is inside `args[0]`?
3. Summation of numbers from arguments
 - ``sum.exe 1 2 3 4 5`` → ``15``
 - ``stoi()`, ``stod()`, ``stoX()`
 - Functions for transformation from **string to** `<something>`
4. **A simple calculator (only for operations +-)**
 - ``calc.exe 1+2+3-4`` → ``2``
 - The previous programs are not needed, but they should give you a lead
 - **Submit the final version to Gitlab → 3 points**
 - **Deadline: Thursday October 9th 5:00**